

U.S. Department of Homeland Security
U.S. Customs and Border Protection



Attachment 01b – Functional Requirements Document

For:
U.S. CUSTOMS AND BORDER PROTECTION
LARGE-SCALE FIXED
NON-INTRUSIVE INSPECTION SYSTEMS

Low-Energy Portal Systems

PREPARED BY:

**OFFICE OF FIELD OPERATIONS
INNOVATION AND STRATEGY DIRECTORATE
NON-INTRUSIVE INSPECTION PROGRAM OFFICE**

December 31, 2025
Version 7.05

WARNING: This document is FOR OFFICIAL USE ONLY. It contains information that may be exempt from public release under the Freedom of Information Action (5 United States Code 522). It is to be controlled, stored, handled, transmitted, distributed, and disposed of in accordance with the Department of Homeland Security policy relating to FOR OFFICIAL USE ONLY information and is not to be released to the public or other personnel who do not have a valid "need-to-know" without prior approval of an authorized Department of Homeland Security official.

Program Signature Page

This document is an authoritative source for the content therein. The approval authorities sign this page.

Prepared by:

Systems Engineering Office
Innovation and Strategy Directorate
Office of Field Operations
U.S. Customs and Border Protection

Reviewed by:

Mindy Sawyer
Technology Acquisition Portfolio
Innovation and Strategy Directorate
Office of Field Operations
U.S. Customs and Border Protection

Approved by:

Darin Asato
Director (A)
Non-Intrusive Inspection Division
Office of Field Operations
U.S. Customs and Border Protection

Date

Revision Summary

Table 1 indicates the changes made to the document since the last release.

Table 1 Revision Table

Version	Date	Location	Description
1.00	March 2020	Washington, DC	Initial Release
2.00	September 2020	Washington, DC	Correct Errata
3.00	February 2023	Washington, DC	Corrected Signatures
4.00	March 2023	Washington, DC	Corrected Formatting
5.00	March 2023	Washington, DC	Corrected Signatures and Formatting
6.00	May 2023	Washington, DC	Updated Signatures
7.00	July 2025	Washington, DC	Updated with changes from lessons learned
7.01	July 2025	Washington, DC	Comments and suggested revisions made by NII Procurement Team
7.02	September 2025	Washington, DC	Revised by NII SE Team
7.03	September 2025	Washington, DC	Comments adjudicated by NII SE Team
7.04	October 2025	Washington, DC	Additional revisions by NII SE Team
7.05	December 2025	Washington, DC	Adjudicated RFI comments

Table of Contents

1	Introduction.....	1
1.1	Mission Needs Statement (MNS) Gaps	2
1.2	System Constraints.....	4
2	LEP Configurations.....	4
3	Requirements	5
3.1	Requirements Definitions	5
3.2	Requirements Numbering	5
4	LEP Use Cases	5
4.1	Inspection Use Case.....	6
5	Functional Requirements	8
5.1	LEP Unit Functional Requirements	9
5.2	LEP Unit X-Ray System Functional Requirements.....	10
5.3	Under-Vehicle Backscatter System (UVBS) Objective – Optional Contract Provision	11
5.4	License Plate Reader (LPR) System Functional Requirements.....	11
5.5	RFID Reader System Functional Requirements	12
5.6	LEP Traffic Control System (TCS) Functional Requirements	12
5.7	Vehicle Detection Functional Requirements.....	13
5.8	CCTV and Camera Systems Functional Requirements	13
5.9	Data Storage (DS) Functional Requirements.....	14
5.10	LEP Work Center Functional Requirements.....	15
5.11	NII Control Center Functional Requirements.....	15
6	Physical Requirements.....	16
6.1	LEP System Physical Requirements	16
6.2	LEP Unit Physical Requirements.....	16
6.3	LEP Unit X-Ray System Physical Requirements	17
6.4	Under-Vehicle Back-Scatter System (UVBS) Physical Requirements.....	17
6.5	LEP Traffic Control System (TCS) Physical Requirements	17
6.6	CCTV and Camera Systems Physical Requirements.....	18
6.7	LEP Work Center Physical Requirements	18
6.8	NII Control Center Physical Requirements	18

7	Performance Requirements	19
7.1	LEP System Performance Requirements	19
7.2	LEP Unit Performance Requirements	19
7.3	LEP Unit X-Ray System Performance Requirements	19
7.4	Under-Vehicle Back-Scatter System (UVBS) - Performance Requirements.....	23
7.5	License Plate Reader (LPR) System Performance Requirements	23
7.6	RFID Reader System Performance Requirements.....	24
7.7	CCTV and Camera Systems Performance Requirements System Performance Requirements.....	24
8	Interface Requirements	24
8.1	LEP System Interface Requirements	24
8.2	LEP X-ray Unit Interface Requirements.....	24
8.3	Under-Vehicle Back-Scatter System (UVBS) Interface Requirements	24
8.4	Traffic Control System (TCS) Interface Requirements	25
8.5	Vehicle Detection Interface Requirements.....	25
8.6	Data Storage (DS) Interface Requirements.....	25
9	Survivability Requirements	25
9.1	LEP System Survivability Requirements.....	25
9.2	LEP Unit Survivability Requirements	26
9.3	Under-Vehicle Back-Scatter System (UVBS) Survivability Requirements	26
9.4	CCTV and Camera Systems Survivability Requirements	26
9.5	LEP Work Center Survivability Requirements	26
9.6	NII Control Center Survivability Requirements	26
10	Human Systems Integration (HSI) Requirements.....	27
10.1	LEP Unit HSI Requirements.....	27
10.2	Work Center HSI Requirements	27
11	Environmental, Safety, and Occupational Health (ESOH) Requirements.....	27
11.1	LEP System ESOH Requirements	27
11.2	LEP Unit X-Ray System ESOH Requirements	29
11.3	Traffic Control System (TCS) ESOH Requirements	29
11.4	CCTV and Camera Systems ESOH Requirements.....	29
12	Interoperability Requirements	30
12.1	LEP System Interoperability Requirements.....	30
12.2	LEP Unit X-Ray System Interoperability Requirements	30

12.3	Under-Vehicle Back-Scatter System (UVBS) Objective - Interoperability Requirements	31
12.4	Vehicle Detection Interoperability Requirements	31
12.5	CCTV and Camera Systems Interoperability Requirements	31
12.6	LEP Work Center Interoperability Requirements	31
13	Security & Privacy Requirements.....	31
13.1	LEP System Security & Privacy Requirements.....	31
13.2	Data Storage (DS) Security & Privacy Requirements	32
13.3	LEP Work Center Security & Privacy Requirements	32
13.4	Access Control Security & Privacy Requirements.....	32
13.5	Auditing Security & Privacy Requirements	33
13.6	Computer Security & Privacy Requirements.....	33
13.7	Identification and Authentication Security & Privacy Requirements.....	35
13.8	Printing Security & Privacy Requirements.....	36
14	Reliability, Availability, and Maintainability (RAM)	36

Table of Tables

Table 1 Revision Table.....	iii
Table 2 NII Mission Needs Statement Gaps.....	2
Table 3 LEP System Functional Components.....	8
Table 4 Detectable Objects	20
Table 5 Concealment Locations	20
Table 6 Transmission Image Quality Indicators (IQI).....	20
Table 7 Backscatter IQI.....	22
Table 8 LEP System Operational Environment.....	27

1 Introduction

United States Customs and Border Protection (CBP) Office of Field Operations (OFO) secures the lawful flow of people and goods into and out of the U.S. CBP Officers must screen all incoming persons and inspect vehicles, conveyances and cargo to detect and resolve potential threats. CBP Officers stand ready to interdict terrorists and their weapons, illicit narcotics and numerous forms of contraband smuggled into the United States. CBP Officers also detect and intercept foreign nationals illegally attempting entry into the United States. These foreign nationals are often persons deemed inadmissible by the Immigration and Nationality Act (INA) and wanted by domestic and/or international law enforcement. In addition, CBP Officers safeguard United States agriculture from harmful animal, fruit, and vegetable diseases and foreign invasive pest species not currently in the United States.

The Low-Energy Portal (LEP) is a drive-through Non-Intrusive Inspection (NII) system utilizing low radiation doses to image an occupied vehicle in motion in pre-primary, primary, and secondary scanning use cases. It provides a detailed image of the vehicle which allows operators to quickly identify anomalies (including explosives, weapons, drugs, and currency). It can be configured to scan multiple lanes of traffic. The LEP systems generally include ancillary system components to assist in identifying the vehicle load or conveyance, including license plate reader (LPR) and Radio Frequency Identification (RFID) systems. Each system shall be capable of providing a NII image with the quality to aid CBP personnel in identifying anomalies (potential contraband) within the conveyance.

CBP is committed to enhancing its mission effectiveness in securing the nation's borders, facilitating legitimate trade, and ensuring efficient travel. Non-Intrusive Inspection (NII) technologies are critical tools in achieving these objectives, providing essential capabilities for detecting illicit goods and threats. To that end, NII intends to modernize its infrastructure to enable CBP to revolutionize how NII data is utilized, including leveraging advanced technologies such as Artificial Intelligence and Machine Learning (AI/ML). All MEP system configurations and their subcomponents shall be capable of exporting all collected image and metadata in real-time to designated CBP IT systems and authorized external platforms. Real-time data export is essential to enable advanced artificial intelligence (AI) technologies to maximize scanner utility, support automated threat detection, and enhance operational efficiency.

The Functional Requirements Document (FRD) details LEP use cases and associated requirements for CBP officers inspecting Privately-Owned Vehicles (POVs) arriving at northern and southern land border Ports of Entry (POEs) throughout the United States, and United States Border Patrol (USBP) checkpoints. Different LEP configurations and concepts of operation (CONOPS) are documented throughout. Each installation will have to be evaluated on case-by-case basis to accommodate the Commercial Off-The-Shelf (COTS) NII technologies based on the physical layout and existing infrastructure. The functional and performance requirements were derived from the NII Operational Requirements Document (ORD) and NII-Integration (NII-I) Consolidate Operational

Requirements Document (CORD). Requirements were further informed by input from CBP officers, lessons learned from previous LEP procurements, analysis, evaluation of various commercial LEP system offerings via technology demonstrations, and prior market research.

This LEP FRD is intended to be issued as Attachment 01 of a larger Statement of Work (SOW) and contracting package. Throughout this document are references to other attachments to the overall SOW.

1.1 Mission Needs Statement (MNS) Gaps

The NII gaps from the NII Mission Needs Statement (MNS)¹ to be addressed by the integrated LEP system are listed in Table 2. Note that the “enhanced” CONOPS discussed in Gaps 4 and 7 are those associated with the NII-I Program and documented in the NII-I Consolidated Operational Requirements Document (CORD).

Table 2 NII Mission Needs Statement Gaps

#	Gap	Description	Type
1	NII System Integration	CBP NII imaging systems are currently not integrated with Radiation Detection Equipment (RDE), the CBP network, data center, local operational command centers nor CBP information technology enforcement applications and databases. Standalone NII large-scale systems do not support automated, efficient data exchange or real-time targeting do not provide a consolidated threat picture for operators and do not support remote performance monitoring and system maintenance. By themselves, Radiation Portal Monitors (RPM) are limited in detecting radiation from well-shielded weapons or radioactive material – such threats must also be scanned for anomalies by an NII imaging system to improve threat detection.	Materiel
2	NII System Standardization	There is a wide variety of NII original equipment manufacturer (OEM) make/model systems currently deployed by CBP which address similar operational requirements, but each requires separate training, integration and maintenance support, needlessly complicating operations and logistics. Use of multiple, different OEM systems results in multiple user interface designs, increased operations and maintenance (O&M) costs and multiple, separate sustainment contracts. System standardization will	Materiel

¹ CBP, “Mission Needs Statement for Non-Intrusive Inspection Program”, signed 14 August 2018.

Large-Scale Fixed Systems

ATTACHMENT 01B – FUNCTIONAL REQUIREMENTS DOCUMENT - LEP

#	Gap	Description	Type
		promote commonality in radiographic imaging formats to lessen the analyst's learning curve with new systems, facilitate centralized analysis with third party systems and lower O&M costs.	
3	Data Networking and Integration	In tandem with #1 and #2, establish a system architecture framework, with interface standards, common data and standard interfaces, and connectivity requirements to support networking and integration of current and future sensors and incremental software improvements.	Materiel
4	Primary NII Image Scanning Capability / Capacity	Currently, NII image scanning is largely a secondary or targeted inspection process. Various operational vectors, including land border Privately Owned Vehicle (POV), truck cargo, seaports, mail, express consignment cargo, and air freight, do not perform NII scanning on a large percentage of incoming traffic / containers. The NII Program must identify and employ new, more efficient technology systems and associated CONOPS, which can offer an opportunity to greatly increase scan rates in these vectors without increasing wait times.	Materiel
7	Adaptation to New Technology and CONOP	Newer Commercial-Off-The-Shelf (COTS) NII technologies have the potential to improve capabilities and greatly increase NII scanning rates using enhanced CONOPS in pre-primary or primary inspection environments. POE facility space, layout and infrastructure are often not designed or built to accommodate these new technologies or enhanced CONOPS.	Materiel Facilities
11	Aging and Obsolete NII Equipment	Many fielded systems, approximately 27 percent of the large-scale and 35 percent of small-scale NII systems, are in need of lifecycle replacement as they have exceeded the vendor defined service life or are already functionally obsolete.	Materiel

#	Gap	Description	Type
15	Chemical, Biological, and Explosive Detection Equipment	There is a lack of NII equipment capacity and capability to detect chemical, biological, and explosive threats and hazards. Additionally, the infrastructure and physical space at many POEs is insufficient to deploy and operate CBE detection systems.	Materiel Facilities

1.2 System Constraints

1. Only active systems with generated X-ray sources will be considered.
2. Per ANSI 43.17-2009 (or latest version), potential radiation dose to vehicle occupants that exceed 25 microrem (μR) per scan will not be considered as stated in Section 11.2.
3. Any wireless communication, including Bluetooth that cannot be disabled by Administrator level operators, will not be considered.
4. Passenger vehicle scanning systems exceeding threshold limits stated in Section 11.2 will not be considered.

2 LEP Configurations

CBP requires three types of low-energy drive-through portals; Type 1: Portals with a limited footprint to support high traffic volume scanning in pre/post-primary passenger vehicle operations; Type 2: Portals with a larger footprint to support secondary passenger vehicle operations; and Type 3: Portals to accommodate bus and empty commercial truck and trailer scanning operations. Each of these types of portals must be configured to meet the variations in Land Port of Entry (LPOE) physical layouts, infrastructure, and operations.

The CBP POEs range from small (one to two lanes) to large (over 30 lanes), requiring a system capable of being configured to meet the needs of the various ports, including in pre-primary, post-primary, secondary, or Border Patrol (BP) checkpoint inspection use cases. Larger LPOEs will require automated systems integrated with the CBP network and information systems to efficiently scan high volumes of traffic. The pre/post-primary systems must be capable of supporting an integrated CBP network CONOPs. Control stations may be single operator workstations networked with a single portal for a small port, or multiple portals multiplexed to multiple workstations in a large port to facilitate a high passenger vehicle throughput

Integration requirements for LEPs are specified by the type of portal. Type 1 portals are pre-primary systems integrated with the CBP network. Type 2 and Type 3 portals are pre-primary or post-primary systems that must be capable of integrating with the CBP network. The Type 2 secondary system is a baseline configuration and must be capable of supporting an integrated CBP network CONOPs.

The requirements for specific types of systems (Type 1, 2, or 3) and configurations are defined in this document. An objective (O) configuration is a pre-primary system integrated with the CBP network and collocated with Radiation Detection Equipment (RDE). These configurations will be implemented incrementally.

3 Requirements

3.1 Requirements Definitions

Threshold (T): The minimum level of operational performance that the Government is willing to accept is considered a threshold value.

Objective (O): An objective is considered a level of performance that significantly improves mission performance, safety, or supportability beyond that of the threshold value, and represents the maximum desired yield for system performance.

- Objective values are not required but are defined to provide guidance to the Contractor with respect to areas where increased capability is of interest to the Government.
- If objectives are defined, the interval between the objective and threshold values for a given parameter is the Government program manager's (PM) trade-space.
- If objective values are listed, the Contractor will evaluate them to determine if they can reasonably be achieved, whether there is a cost or schedule impact, and the extent of the impact. The impact of achieving objective values will be brought to the attention of the Government for review.

Requirement Statement: This is a threshold requirement that contains a single parameter value or function.

3.2 Requirements Numbering

Each requirement in this FRD is assigned a unique ID number of the form of XX.YY.ZZ.

- XX indicates the requirement category number.
- YY indicates the component number.
- ZZ defines the individual requirement number.

4 LEP Use Cases

The LEP passenger vehicle scanning system will be used in pre-primary and post-primary CONOPs. The post-primary CONOPs is used in cases where the physical layout of the POE does not allow positioning of the LEP ahead of the primary booths (e.g., primary is too close to the border). Post-primary also provides an additional inspection tool that can be used to augment physical inspections.

The pre-primary CONOPs utilizes technology to scan passenger vehicles prior to arriving at the primary booth providing the Primary Officer with an additional decision aid. This pre-primary CONOPs will be performing scanning and adjudicating activities prior to the passenger vehicle arriving at the primary point of inspection into the United States. This drive-through CONOPs will increase throughput enabling a higher scanning percentage of vehicles entering the U.S. and thereby increasing the probability of detecting illegal contraband entering the country to deter smuggling via passenger vehicles.

4.1 Inspection Use Case

The use case describes LPOE CBP operations when scanning inbound passenger vehicles at a land border crossing. Steps on CBP systems external to the passenger vehicle scanning system are included to provide context to the operation and to identify interfaces to CBP systems.

Use Case Pre-Condition:

1. The LEP system is ready for operation.
2. A vehicle is arriving at a POE.

Use Case Post Condition:

1. The vehicle is cleared and proceeds to exit or is referred for secondary inspection.
2. The Data Store (DS) saves the data package.

Use Case Steps

1. This system confirms the vehicle is not oversized prior to the vehicle proceeding.
2. If the vehicle chooses to opt-out of scanning, then the vehicle proceeds to the opt-out lane.
3. The system's pre-primary License Plate Reader (LPR) captures the front license plate (LP) and front image of the subject vehicle. Note: Not all vehicles are required to have front LP.
4. The Traffic Control Indicator signals the subject vehicle to enter the passenger vehicle lane.
5. The Traffic Control Indicator signals any following vehicles to stop until the portal is clear to proceed.
6. The system captures driver and passenger(s) RFID.
7. The vehicle advances through the passenger vehicle lane and triggers the vehicle presence detection.
8. The vehicle presence detection activates vehicle scanning and starts capturing the X-ray imagery.

9. The subject vehicle continues to move through the passenger vehicle lane at approximately 5 mph.
10. If the subject vehicle advances at a speed that could potentially expose vehicle occupants to radiation dose above 25 μ R per scan, the system stops scanning.
11. If scan stops due to a slow speed (Step 10), then the data package is annotated that the “scan stopped due to slow speed.”
12. When the system detects the end of the subject vehicle:
13. System stops scanning
14. System’s LPR captures the rear LP and image.
15. The subject vehicle proceeds to next available primary lane.
16. When the subject vehicle clears the last lane sensor, the Traffic Control Indicator signals the next vehicle to enter system screening portal.
17. The LEP System creates and timestamps the data package and assigns a unique case record number. LEP System sends the data package to the Data Storage (DS).
18. The LEP data packages plus other LEP health and performance data are transmitted to the CBP network and data center/cloud.
19. The LEP data package is ‘posted’ to the LEP queue in the control booth / NII control center.
20. A LEP (Work Center) operator/analyst is automatically assigned a LEP scan image in the order scanned, which triggers the analysis process.
21. The LEP operator/analyst provides a ‘red’ suspect referral or ‘green’ non-suspect release recommendation and annotates the preliminary decision in the Government system. The LEP operator/analyst annotates the image with overlay graphics to indicate the location of the anomaly if present.
22. If ‘red,’ the LEP scan image is available to officers/agents for reference to perform follow on enforcement actions.
23. The POV pulls up to the Primary booth.
24. The Primary officer observes the vehicle, reviews documentation, interacts with the driver and passenger(s). The officer also checks the red/green referral recommendation from the image analysis. The Primary officer makes a decision to either refer the POV for further inspection (i.e., Secondary) or allow entry into the U.S.
25. The Primary officer inputs the decision into the Government system.
26. The Primary officer either directs the POV to proceed to the exit gate, or Secondary inspection area for further investigation.

27. The use case ends when the POV either proceeds to the POE exit or is detained for further inspection.

5 Functional Requirements

Table 3 list the major components of the LEP system. Besides these major components the system also includes additional associated hardware and software elements. Note that to keep the numbering consistent with the Multi-Energy Portal (MEP) FRD, some components that are listed in the table are Not Applicable (N/A) to the LEP.

Table 3 LEP System Functional Components

Component	Functional Component Description
1. LEP Unit	The LEP Unit refers to the systems and subsystems, which are installed in one traffic lane to execute the full scanning operation. The LEP system is a drive-through, low-energy X-ray system. The LEP system shall include integrated RFID transponder reader and LPR for use at LPOE, to uniquely identify the LEP data package for the specific vehicle/container being scanned. The LPOE drive-through systems may include as options an integrated Backscatter or Under Vehicle Back-Scatter system (UVBS).
2. LEP Unit X-Ray System	X-ray scanning system capable of providing images and material characterization of the POV, occupants, and contents. As an option, the LEP Unit X-ray may include a backscatter X-ray system (in addition to primary transmission X-ray system).
3. Under-Vehicle Back-Scatter System (UVBS) - Objective – Optional Contract Provision	Backscatter X-ray scanning system capable of providing images of the undercarriage of vehicles.
4. License Plate Reader (LPR) System (Land Border configuration)	Identification system that uses OCR technology to read the front and rear license plates of the POV. Provides jurisdiction (e.g., state, province) and plate alphanumeric ID.
5. RFID Reader System (Land Border configuration)	Identification system that uses RF signals to read passive (backscatter) transponders (tags) of the driver and/or passenger(s) if enrolled a trusted partner program (e.g., NEXUS, SENTRI) that provides expedited screening and clearance.
6. Optical Character Recognition (OCR) System	N/A
7. Check-in or Staging Kiosk	N/A
8. Traffic Control System (TCS)	The Traffic Control System (TCS) will ensure the efficient and effective flow of traffic without negatively impacting the overall flow of vehicles through the POE. This capability may include lights, signage, control arms, pavement striping, lane barriers, and other elements as required (to the extent possible, these elements should be customer serviceable). The TCS shall

Component	Functional Component Description
	be integrated with LEP operation. It will include a traffic/lane controller to automatically and actively guide traffic through the system steps and manage lane closures, LEP system outages, etc., with operator or smart queuing system input.
9. Vehicle Detection	System used to detect the presence of vehicles in a specific position in a lane. Examples include inground inductive loops or break-beam sensors.
10. Closed-Circuit Television (CCTV) and Camera Systems	Closed-circuit television (CCTV) systems used for surveillance. Has the capability to record and view video. Uses pan-tilt-zoom (PTZ) video cameras.
11. Data Storage (DS)	Local computer data storage to preserve images and data packages.
12. LEP Work Center	Workstations where CBP officers and agents can view and enhance (e.g., zoom, contrast) X-ray images to look for the presence of anomalies that would warrant further inspection of vehicles or cargo.
13. NII Control Center	The NII Control Center allows for remote operational control of the LEP System. Remote control includes the operation of the LEP Unit(s), the traffic control system, required communications to support the workflow, situational awareness (CCTV system displays), and any other requirements to fully operate the LEP System and perform analysis of LEP data. In some secondary locations, the control center can be a simple operator / analyst booth located adjacent to the LEP system. In other applications, it can be a stand-alone facility or dedicated office space within an existing POE or regional facility. Within the control center, there can be one or more work centers, consisting of LEP operator / analysis workstations and an associated CBP networked workstation hosting law enforcement applications such as ACE-TMM, TECS, etc.

5.1 LEP Unit Functional Requirements

The LEP Unit refers to the systems and subsystems which are installed in one traffic lane or one inspection/scanning area to make up the full scanning operation.

- 5.1.1 The LEP Unit shall be designed and configured for operation from the LEP Work Center or NII Control Center.
- 5.1.2 For Type 1 and Type 2 systems, the LEP Unit, including all associated equipment, shall accommodate vehicles up to 9' 0" in height (T)
- 5.1.3 For Type 1 and Type 2 systems, the LEP Unit, including all associated equipment, shall have a tunnel width that is at least 9' 0" wide (T).
- 5.1.4 For Type 3 systems, the LEP Unit, including all associated equipment, shall accommodate vehicles up to 14' 6" in height (T)

- 5.1.5 For Type 3 systems, the LEP Unit, including all associated equipment, and shall have a tunnel width that is at least 12' 0" wide (T).
- 5.1.6 For Type 3 systems, the LEP Unit, including all associated equipment, should accommodate vehicles up to 15' 0" in height (O).

5.2 LEP Unit X-Ray System Functional Requirements

- 5.2.1 The LEP unit shall use active X-Ray technology.
- 5.2.2 For Type 1 systems, the LEP unit shall collect a top-down or bottom-up transmission image (T) Top-down is preferred.
- 5.2.3 For Type 1 systems, the LEP unit should collect a backscatter image (O).
- 5.2.4 The LEP unit shall start scanning when the vehicle enters the portal.
- 5.2.5 The LEP unit shall stop scanning when the subject vehicle exits the scanning area.
- 5.2.6 The LEP unit shall display the entire length of the vehicle (front to end of last trailer), from the ground to the roof for the side views and the complete length and width of the vehicle for the top-down view or bottom-up view.
- 5.2.7 The LEP Unit shall be capable of material discrimination using the multi-energy capability to distinguish, at a minimum, between organic (low Z)², mix of inorganic/organic (intermediate Z) and inorganic (high Z) materials.
- 5.2.8 The LEP Unit shall allow the operator-analyst to toggle material discrimination colors on and off.
- 5.2.9 The LEP Unit shall distinguish between low ($Z \leq 10$), intermediate ($10 < Z < 20$), and high ($Z \geq 20$) materials using different colors for their display. Material categories are to be color-coded as follows: low (orange), intermediate (green) and high (blue) (T). Shades of these colors are to be used to indicate thickness. All colors should be in compliance with the Unified File Format, Version 3.
- 5.2.10 The LEP Unit should distinguish Heavy Metals ($Z \geq 30$) (Color not defined by CBP) (O).

² Z = atomic number or nuclear charge number

- 5.2.11 The Type 2 and 3 systems shall display six images of the conveyance using both transmission and backscatter technologies: top-down, both sides in transmission and backscatter.

5.3 Under-Vehicle Backscatter System (UVBS) Objective – Optional Contract Provision

- 5.3.1 If included, the UVBS backscatter X-ray shall be integrated into the scanning operation of the LEP Unit to perform high resolution scan of the undercarriage of a standard width vehicle and variable under-vehicle heights.
- 5.3.2 If included, the UVBS shall automatically begin operation when the vehicle triggers the system that the scanning run has begun.
- 5.3.3 If included, the UVBS shall stop capturing undercarriage imagery when the subject vehicle exits the scanning area.
- 5.3.4 If included, the UVBS scan image shall be spatially aligned with the LEP X-ray (transmission / backscatter) scan images.
- 5.3.5 If included, the UVBS system shall include an automated means to be kept free of obstructions (dust/debris) at all times to ensure the acquisition of X-ray image for each inspection.
- 5.3.6 If included, the UVBS system shall capture the complete undercarriage.

5.4 License Plate Reader (LPR) System Functional Requirements

- 5.4.1 The LPR system shall be capable of reading a single front license plate and a single rear license plate, where the license plates comply with American Association of Motor Vehicle Administrators (AAMVA) License Plate Standard, Edition 3 (or latest standard)
- 5.4.2 The LPR system shall provide a broad image of the front and rear areas of the vehicle, which can be used to confirm the accuracy of the optical characters.
- 5.4.3 The LPR shall create an alert if the front license plate is missing or not detected. Note that some jurisdictions do not require front plates
- 5.4.4 The LPR shall validate that the rear license plate matches the front license plate.
- 5.4.5 The LPR shall create an alert if the front and rear license plates do not match.

5.5 RFID Reader System Functional Requirements

- 5.5.1 The RFID system shall be capable of reading GEN 2 RFID tags, per ISO-18000-63 version 2 specification, Information Technology - Radio Frequency Identification for Item Management - Part 63 with EPC Global / GS 1 GDTI-96 tag format.
- 5.5.2 The RFID system shall filter out non-DTOPS tags based on CBP defined header in memory bank #1 EPC Global / DTOPS registration number.

5.6 LEP Traffic Control System (TCS) Functional Requirements

- 5.6.1 The TCS shall ensure the efficient and effective flow of traffic without negatively impacting the overall flow of vehicles through the POE.
- 5.6.2 The TCS shall include positive control of each conveyance before entering the LEP lane. Positive control of the traffic system is the point at which a driver is forced to proceed through the LEP system while being directed by lights, signage, control arms, etc.
- 5.6.3 The TCS shall control travel of the conveyance through the LEP lane until exiting.
- 5.6.4 The TCS shall control the area after the scan to allow the conveyance the proper stopping distance including the post-LEP staging lane area, as applicable.
- 5.6.5 If required due to traffic congestion, the TCS shall allow the vehicle to safely come to a stop after fully exiting the LEP Unit.
- 5.6.6 The TCS shall not require the vehicle to stop after the scanning sequence is started.
- 5.6.7 The TCS shall be capable of directing traffic for driver opt-outs and oversize vehicles.
- 5.6.8 The TCS shall control traffic flow when exception conditions occur. Note: Examples of exception conditions are: Lane Closure (e.g. disabled vehicle); Response to emergency conditions either by CBP Officers or by other emergency services.
- 5.6.9 The LEP TCS shall be configured to allow vehicles to pass through the LEP Unit if the LEP Unit is disabled.
- 5.6.10 The TCS shall be configured to include pre-set, automatic traffic pattern sequences to accommodate standard and non-standard traffic patterns and exception conditions (e.g. All lanes open, reduced number of lanes, specific lane closes, etc.) initiated by CBP officers in the NII Control Center.

- 5.6.11 The TCS shall be designed and configured for unmanned operations during standard operating scenarios.
- 5.6.12 The TCS shall be designed and configured for unmanned operations for identifiable non-standard operating conditions such as a LEP Unit not being available for scanning.
- 5.6.13 The TCS shall verify that the vehicle is not oversized (height and width) prior to arriving at the Traffic Control Indicator using mechanical or electronic means.
- 5.6.14 The TCS shall take inputs from lane capacity at the primary booth and other sensors to automatically assign trucks to LEP lanes, bypass lanes, or specific booths to maximize throughput and maintain optimal primary operations.
- 5.6.15 The TCS shall optimize traffic flow for all operational LEP lanes.
- 5.6.16 The TCS shall include logic-based routing to optimize traffic flow.
- 5.6.17 The TCS shall be configured to maximize throughput for all operational LEP lanes and provide CBP Officers with the ability to actively route traffic past the LEPs when necessary to maintain optimal primary operations.

5.7 Vehicle Detection Functional Requirements

- 5.7.1 The LEP Unit(s) shall detect the presence (or absence) of vehicles in the LEP lane.
- 5.7.2 The LEP Unit(s) shall measure the vehicle's instantaneous speed to the nearest 0.1 MPH from 0 to 15 MPH as it approaches and passes through each LEP Unit.

5.8 CCTV and Camera Systems Functional Requirements

- 5.8.1 The CCTV and camera systems shall provide a Fixed CCTV system for each LEP Unit.
- 5.8.2 The Fixed CCTV system shall capture live video of vehicles traveling through the LEP Units.
- 5.8.3 The Fixed CCTV system shall provide full 360-degree coverage of the LEP Unit operating area.
- 5.8.4 The CCTV and camera systems shall provide PTZ cameras for each LEP Unit.
- 5.8.5 PTZ cameras positioned at the entrance and exit of each LEP Unit shall provide sufficient resolution and image quality to permit qualitative descriptions (make, model, color and length) of all and vehicles that pass through the LEP.

Note: The CBP Border Security Deployment Program (BSDP) has established an approved specification for surveillance video cameras and recording called CAVSS. Any camera covering interactions between CBP officers and drivers must comply with CAVSS requirements.

- 5.8.6 The CCTV and camera systems shall provide elevated surveillance PTZ cameras for wide-area situational awareness
- 5.8.7 The surveillance PTZ cameras shall have the capability to focus using pre-programmed presets.
- 5.8.8 All camera views (CCTV, LEP PTZ, surveillance PTZ) shall be automatically recorded.
- 5.8.9 The CCTV and camera systems shall be capable of providing screenshots at any time.

5.9 Data Storage (DS) Functional Requirements

- 5.9.1 The LEP System Data Storage (DS) shall store the data packages (including X-ray images, IDs, and metadata) sent to the NII Control Center or Cloud.
- 5.9.2 The LEP System DS shall store X-Ray images in an uncompressed data format.
- 5.9.3 The LEP System DS shall retain raw X-Ray image data for each data package in addition to any proprietary / processed X-Ray image format.
- 5.9.4 The LEP System DS stored images shall retain the full dynamic gray-level range of the X-Ray image
- 5.9.5 The LEP System DS shall store data packages for a rolling first-in first-out period of thirty (30) days.
- 5.9.6 The LEP System DS shall support operator-defined queries, including standard reports for scans performed, scans per day / per analyst and other Government defined metrics
- 5.9.7 The LEP System shall provide an option at the Administrator level for auto delete of files that are over 30 days old.
- 5.9.8 The LEP System DS shall have a full capacity backup system that is configured to automatically backup files at designated intervals
- 5.9.9 The LEP System DS shall delete oldest data packages to restore room for new packages when storage reaches 99% of its maximum storage capacity.
- 5.9.10 The LEP System DS shall not over-write any previous decisions or annotations made by any analyst due to any analyst

manipulations or actions at any System workstations, regardless of location.

5.10 LEP Work Center Functional Requirements

- 5.10.1 The LEP Work Center(s) shall allow all the required functions to fully adjudicate the X-ray images.
- 5.10.2 The LEP Work Center(s) shall provide the ability to do the following: enhance the image(s), change the magnification, select different color palettes, optimize the image's contrast and brightness, sharpen the image, highlight potential areas of interest, annotate the file for secondary inspection, input decisions on shipment adjudication and perform other standard and required operations to meet the objective of detecting contraband and anomalies.
- 5.10.3 The LEP Work Center(s) shall display complete images free of vertical or horizontal lines, any noise and any data or corruption that may confuse or hinder the operators' display, i.e., missing or incomplete data such as partial vehicle scans. This requirement is intended to present images of the entire vehicle, free of noise and garbage.
- 5.10.4 The System shall display the username and local time with the image and the data package.
- 5.10.5 The System shall tag each scan analysis with the username who conducted it.

5.11 NII Control Center Functional Requirements

- 5.11.1 Supervisors in the NII Control Center with Administrator accounts shall remotely control the LEP System.

Note: Remote control includes full remote operations for the system including the operation of the LEP Unit, NII Control Center Situational Wall (SW), the traffic control system, any required communications to support the work flow, situational awareness (CCTV system displays), and any other requirements to fully operate the LEP System.

- 5.11.2 The NII Control Center SW shall include two (2) scan Queues on separate monitors as described below: a) Primary scan queue shall be displayed on a screen for the un-adjudicated scans from the LEP Units. b) Secondary scan queue shall be displayed on a screen for all shipments which are 'red lighted' and referred for secondary examination.

- 5.11.3 For each LEP Unit, the NII Control Center SW video wall shall allow display for each CCTV / PTZ system so that the images associated with each LEP unit (lane) are easily identified.

6 Physical Requirements

Physical requirements include those covering components, dimensions, and installation.

6.1 LEP System Physical Requirements

- 6.1.1 The LEP System shall include computers and server(s) with enough memory, speed and other features to meet all the requirements of this FRD and supporting contract SOW / references.
- 6.1.2 Printers shall be capable of color printing

6.2 LEP Unit Physical Requirements

- 6.2.1 The LEP Unit shall operate whether installed in a pre-primary, post-primary or secondary position.
- 6.2.2 For Type 1 systems, the LEP Unit shall be able to be installed within a defined footprint no wider than 14 feet and no longer than 20 feet and no higher than 15 feet (T). These dimensions do not include any fencing or shielding walls and other auxiliary equipment or exit distance.
- 6.2.3 For Type 2 systems, the LEP Unit shall be able to be installed within a defined footprint no wider than 24 feet and no longer than 24 feet and no higher than 15 feet (T). These dimensions do not include any fencing or shielding walls and other auxiliary equipment or exit distance.
- 6.2.4 For Type 2 systems, the LEP Unit should be able to be installed within a defined footprint no wider than 16 feet and no longer than 20 feet and no higher than 15 feet (O). These dimensions do not include any fencing or shielding walls and other auxiliary equipment or exit distance.
- 6.2.5 For Type 3 systems, the LEP Unit shall be able to be installed within a defined footprint no wider than 29 feet and no longer than 27 feet and no higher than 21.5 feet (T). These dimensions do not include any fencing or shielding walls and other auxiliary equipment or exit distance.
- 6.2.6 For Type 3 systems, the LEP Unit should be able to be installed within a defined footprint no wider than 20 feet and no longer than 22 feet and no higher than 21.5 feet (O). These dimensions do not include any fencing or shielding walls and other auxiliary equipment or exit distance.

6.3 LEP Unit X-Ray System Physical Requirements

- 6.3.1 For Type 2 and 3 systems, the side view shall include from the ground to the roof without ramps or raised platforms.

6.4 Under-Vehicle Back-Scatter System (UVBS) Physical Requirements

- 6.4.1 If included, the UVBS shall be flush mounted with the road.
- 6.4.2 If included, the UVBS physical dimensions (length, width, depth) shall not cause an increase LEP lane dimensions beyond what is stated in Section 6.3.

6.5 LEP Traffic Control System (TCS) Physical Requirements

- 6.5.1 The TCS shall cover all lanes leading into or bypassing the LEP Units and to the agreed upon distances before and after the LEP Units.
- 6.5.2 As applicable, the TCS controller shall be centrally located in the NII Control Center with a separate operating workstation.
- 6.5.3 The TCS shall be located and configured to allow a vehicle to achieve required minimal scan speed prior to entering the LEP Unit scan tunnel.
- 6.5.4 The TCS should achieve the traffic control goals without the use of gate arms (O).
- 6.5.5 The TCS shall have mechanical barriers to restrict oversized conveyances defined by height and/or width (6.2.2 to 6.2.6) prior to entering the LEP lane to prevent collision damage to the LEP Unit.
- 6.5.6 The TCS shall include active signage for all lanes to direct vehicles to open LEP lanes or past the LEP lanes when necessary.
- 6.5.7 The TCS shall include active signage for all lanes after the LEP to direct vehicles to open primary lanes.
- 6.5.8 The TCS shall include a minimum of one bypass lane for every three LEP lanes to ensure sufficient bypass capacity.
- 6.5.9 The TCS shall include license plate readers.
- 6.5.10 The TCS shall have sufficient infrastructure (power and fiber) to support additional technologies, such as facial recognition cameras, touch screen monitors, and other future systems. Refer to the SOW for additional details.
- 6.5.11 The TCS components shall be physically installed in locations that ensure requirements in Section 5.6.2 are met.

6.6 CCTV and Camera Systems Physical Requirements

- 6.6.1 The Fixed CCTV system shall be a high-definition (HD) or greater CCTV for each of the LEP Units, and associated operations, that is compliant with CBP CAVSS and approved by CBP BSDP.
- 6.6.2 The LEP Unit PTZ cameras shall be placed to allow for viewing of the license plates of the vehicle and provide enough range to view the vehicle and occupants.
- 6.6.3 The LEP Unit PTZ cameras shall include cameras at the entrance and exit of each LEP Unit
- 6.6.4 The Elevated PTZ cameras shall be attached to one or more LEP Units to provide for full situational awareness of the traffic lanes leading up to the LEP Units and the lane areas between the LEP Units and the Primary Booths.
- 6.6.5 The CCTV and Camera System shall include color monitor(s).
- 6.6.6 The color monitors shall have HD or greater resolution
- 6.6.7 The color monitors shall be 32" or larger.
- 6.6.8 The LEP System shall include Network Video Recorder (NVR) with a minimum 30-day recording capability capable of playing back any camera view.

6.7 LEP Work Center Physical Requirements

- 6.7.1 The LEP workstations shall comply with DHS Instruction 119-02-03, "DHS Workspace Standard"
- 6.7.2 The LEP workstation monitors (excluding the Government workstation) shall be a minimum of 40" in size or the images may be compiled on a large screen (50+") which provides images of the same equivalent size.

6.8 NII Control Center Physical Requirements

- 6.8.1 The NII Control Center shall include an administrative work space, where a supervisor with Administrator account can manage workload situations, emergencies, respond to system alarms (e.g. gate-arm crash), activate and deactivate LEP lanes, effecting an orderly shutdown in case of power failures, etc.
- 6.8.2 The NII Control Center shall comply with DHS Instruction 119-02-03, "DHS Workspace Standard"; and 2023 CBP LOPE Design Standards and CBP security requirements and CBP OIT requirements.
- 6.8.3 The NII Control Center SW shall include one (1) HD or greater resolution screen TV for monitoring news and weather channels.

7 Performance Requirements

General performance relates to quantity, speed, and other general performance characteristics of the system.

7.1 LEP System Performance Requirements

- 7.1.1 The LEP System data shall be available to the LEP Work Center in five (5) seconds (T) after the truck scan is complete.
- 7.1.2 The LEP System data should be available to the LEP Work Center in three (3) seconds after the truck scan is complete (O).
- 7.1.3 The LEP System data shall begin to upload to the CBP Network in two (2) seconds (T) after the truck scan is complete.
- 7.1.4 The LEP System data should be available to the LEP Work Center or NII Cloud in one (1) second after the truck scan is complete (O).

7.2 LEP Unit Performance Requirements

- 7.2.1 The LEP Unit shall scan conveyances traveling with speeds of 2 to 10 MPH.
- 7.2.2 For Type 1 pre/post primary systems, the LEP Unit shall have a minimum throughput of 100 (T) vehicles per hour.
- 7.2.3 For Type 2 and 3 secondary and bus systems, the LEP Unit shall have a minimum throughput of 20 (T) vehicles per hour.

7.3 LEP Unit X-Ray System Performance Requirements

- 7.3.1 The LEP system shall display radiographic images free of vertical or horizontal lines or any noise, and not missing any data that may hinder the operators clear view of the message (i.e., missing or incomplete data) such as partial vehicle scans.
- 7.3.2 The system shall display radiographic images of sufficient quality to enable a trained CBP officer to identify the objects described in Table 4 Detectable Objects concealed in any of the locations listed in Table 5 Concealment Locations with a minimum success rate of 50 (T) percent.
- 7.3.3 The system should display radiographic images with the quality for a system trained CBP officer to discriminate the object described in Table 4 Detectable Objects concealed in any of the locations listed in Table 5 Concealment Locations with at least 95 (O) percent success.

Table 4 Detectable Objects

Item	Description	Dimensions	Density
1	Narcotic	8" x 5.5" x 1.5"	0.74 to 1.5 g/cm ³
2	Handgun	185 mm x 128 mm (7.28" x 5.04")	7.8 g/cc ³

Table 5 Concealment Locations

Item	Location	System Type 1, 2	System Type 3
1	Doors	X	
2	Rear Seat	X	Passenger Seats
3	Fuel Tank	X	X

Table 6 is applicable to Type 1, 2, or 3 systems.

Table 6 Transmission Image Quality Indicators (IQI)

Requirement	Type 1	Type 2	Type 3
7.2.5 The system shall penetrate X (mm) of steel in accordance with test procedure in ANSI 42-46-2008.	19 (T)	19 (T)	19 (T)
7.2.6 The system should penetrate X (mm) of steel in accordance with test procedure in ANSI 42-46-2008.	65 (O)	65 (O)	65 (O)
7.2.7 The system's radiographic images shall discriminate object to background density by at least X (%) in accordance with test procedure in ANSI 42-46-2008.	5% (T)	5% (T)	5% (T)

Requirement	Type 1	Type 2	Type 3
7.2.8 The system's radiographic images should discriminate object to background density by at least X (%) in accordance with test procedure in ANSI 42-46-2008.	3% (O)	3% (O)	3% (O)
7.2.9 The system's horizontal spatial resolution shall be less than X (mm) in accordance with test procedure in ANSI 42-46-2008.	6 (T)	9 (T)	10 (T)
7.2.10 The system's horizontal spatial resolution shall be less than X (mm) in accordance with test procedure in ANSI 42-46-2008.	2 (O)	6 (O)	6 (O)
7.2.11 The system's vertical spatial resolution shall be less than X (mm) in accordance with test procedure in ANSI 42-46-2008.	8 (T)	12 (T)	20 (T)
7.2.12 The system's vertical spatial resolution shall be less than X (mm) in accordance with test procedure in ANSI 42-46-2008.	2 (O)	8 (O)	12 (O)

Table 7 is applicable to Type 1 system with backscatter technology and Type 2 and 3 systems.

Table 7 Backscatter IQI

Requirement	Type 1	Type 2	Type 3
7.2.13 The system shall penetrate X (mm) of steel in accordance with test procedure in ANSI 42-46-2008.	6 (T)	6 (T)	6 (T)
7.2.14 The system shall penetrate X (mm) of steel in accordance with test procedure in ANSI 42-46-2008.	8 (O)	8 (O)	8 (O)
7.2.15 The system's horizontal spatial resolution shall be less than X (mm) in accordance with test procedure in ANSI 42-46-2008.	6 (T)	9 (T)	10 (T)
7.2.16 The system's horizontal spatial resolution shall be less than X (mm) in accordance with test procedure in ANSI 42-46-2008.	4 (O)	6 (O)	6 (O)
7.2.17 The system's vertical spatial resolution shall be less than X (mm) in accordance with test procedure in ANSI 42-46-2008.	9 (T)	10 (T)	12 (T)
7.2.18 The system's vertical spatial resolution shall be less than X (mm) in accordance with test procedure in ANSI 42-46-2008.	4 (O)	4 (O)	4(O)
7.2.19 The System radiographic images shall provide a layer contrast of at least X (mm) in accordance with test procedure in ANSI 42-46-2008.	1.5 (T)	2 (T)	2.5 (T)

Requirement	Type 1	Type 2	Type 3
7.2.20 The System radiographic images shall provide a layer contrast of at least X (mm) in accordance with test procedure in ANSI 42-46-2008.	1.0 (O)	1.5 (O)	2 (O)
7.2.21 The System radiographic images shall provide an isolation contrast of at least X (mm) in accordance with test procedure in ANSI 42-46-2008.	1 (T)	1 (T)	1(T)

7.3.4 The LEP Unit shall support display of radiographic images at 4096 (T) gray levels or greater.

7.3.5 The LEP Unit should support display of radiographic images at 65,536 (O) gray levels or greater.

7.4 Under-Vehicle Back-Scatter System (UVBS) - Performance Requirements

7.4.1 If included, the UVBS scan speed shall be equal to or greater than the maximum LEP Unit scan speed.

7.4.2 If included, the UVBS shall meet the backscatter imaging requirements in Table 7.

7.5 License Plate Reader (LPR) System Performance Requirements

7.5.1 The LPR camera capture rate for image shall be 99.5% (T). Note: "Capture" is defined as recording a human-readable image of the license plate.

7.5.2 The LPR camera capture rate for image should be 100% (O). Note:

7.5.3 The LPR shall read license plates with a 90.0 (T) percent accuracy of the plate alphanumeric identification and the jurisdiction in all lighting conditions (including day, night, and twilight) for all 50 U.S. states, 31 Mexican states and 12 Canadian provinces/territories.

Note: Exclusions for physical obstruction or damage are not allowed.

7.6 RFID Reader System Performance Requirements

- 7.6.1 The LEP RFID system shall read the RFID tags (e.g., SENTRI, NEXUS) with a 99% (T) accuracy for readable tags.
- 7.6.2 The LEP RFID system should read the RFID tags with 100% (O) accuracy for readable tags.
- 7.6.3 RFID reads of tags in other than the correct lane (colloquially cross-lane reads) shall be less than 0.2% (T) of all RFID reads.
Note: Where more than one LEP system is installed, or other RFID readers are installed in proximity.

7.7 CCTV and Camera Systems Performance Requirements System Performance Requirements

- 7.7.1 Area surveillance component (beyond LEP lanes) shall provide a recording capacity of 30 calendar days for the retrieval of video.

8 Interface Requirements

The interface requirements for the LEP to the NII Control Center and Cloud are described here.

8.1 LEP System Interface Requirements

- 8.1.1 The LEP System shall interface to the NII Control Center, (future) Cloud, and (future) NII-I system in accordance with Attachment 02 – Integration - Interface Control Document (ICD) of the SOW.
- 8.1.2 The LEP System shall support (future) interface to Government network and Cloud storage of all LEP data elements including image and metadata, health, performance, fault and configuration status.
- 8.1.3 The LEP System shall include LPR, and/or RFID metadata as a means to identify a particular vehicle, and to associate it with its NII image package as the vehicle proceeds through the POE scanning and inspection process.

8.2 LEP X-ray Unit Interface Requirements

- 8.2.1 The System shall interface to any RPMs in proximity to enable RPM blanking.

8.3 Under-Vehicle Back-Scatter System (UVBS) Interface Requirements

- 8.3.1 If included, the UVBS images shall be integrated into the data package.

8.4 Traffic Control System (TCS) Interface Requirements

- 8.4.1 The TCS traffic lights shall include a red light representing stop and green light as the indicator to proceed.
- 8.4.2 Gate arms (if required) shall provide an alarm to the control center LEP operator if they reverse on contact with an obstruction.
- 8.4.3 The TCS shall include mechanical indicators to warn and divert drivers of oversized vehicles which could damage LEP Unit prior to entering the LEP lane and be placed to allow sufficient maneuvering space for the truck to avoid entering the LEP.
- 8.4.4 The TCS shall include active signage to direct vehicles to open MEP lanes as well as past the MEP lanes when necessary.

8.5 Vehicle Detection Interface Requirements

- 8.5.1 The Vehicle Detection System shall provide a visual cue to the driver communicating his/her speed and whether they are within the correct speed range for effective scanning.

8.6 Data Storage (DS) Interface Requirements

- 8.6.1 The LEP System shall export data packages in accordance with *Attachment 02 – Integration* - Interface Control Document (ICD) of the SOW.
- 8.6.2 Locally stored files, records, and scan images shall be accessible to Administrators who are able to copy to removable media storage devices: encrypted, FIPS 1402 Level 2 or greater and FIPS 197 compliant flash and hard drives.
- 8.6.3 The LEP System DS shall create a warning when storage reaches 70% of its maximum storage capacity.
- 8.6.4 The LEP System DS shall create an alert when storage reaches 90% of its maximum storage capacity
- 8.6.5 The LEP System DS shall create an alarm when storage reaches 99% of its maximum storage capacity.

9 Survivability Requirements

Survivability requirements describe the capability of the system to fulfill its mission in a timely manner in the presence of attacks, failures, or accidents.

9.1 LEP System Survivability Requirements

- 9.1.1 The LEP System shall include hardware to protect from lighting strikes or power surges.

- 9.1.2 The LEP System shall condition incoming power to within +10%.
- 9.1.3 The LEP System shall have backup capability to maintain power sufficient to allow thirty (30) minutes of standard operation and graceful shut down of systems.
- 9.1.4 The LEP System exterior components and housings shall be high strength, anti-corrosive, and weathertight for the full design life of 10+ years.

9.2 LEP Unit Survivability Requirements

- 9.2.1 The LEP Unit shall have a Power Conditioner and UPS, Online Double Conversion type, to protect the system from electrical power degradation / fluctuations / loss.

9.3 Under-Vehicle Back-Scatter System (UVBS) Survivability Requirements

- 9.3.1 If included, the UVBS shall have provision for preventing water intrusion or otherwise manage water which infiltrates the unit to prevent damage / failure of the unit.
- 9.3.2 If included, the UVBS housings shall be high strength, anti-corrosive, and weather tight for the full design life of 10+ years.
- 9.3.3 If included, the UVBS shall survive vehicles that weigh up to 26,000 lbs.

9.4 CCTV and Camera Systems Survivability Requirements

- 9.4.1 Backup power shall be provided for all surveillance system components for (one) 1 hour of operation.

9.5 LEP Work Center Survivability Requirements

- 9.5.1 System shall include an UPS, Online Double Conversion type, for the LEP Work Center and a power conditioner to regulate and maintain power sufficient to allow thirty (30) minutes of standard operation and a graceful shut down of systems.

9.6 NII Control Center Survivability Requirements

- 9.6.1 The NII Control Center shall be connected into a backup generator system capable of running all direct and support functions of the NII Control Center for a minimum period of 24 hours.
- 9.6.2 The NII Control Center shall have redundant backup system requirements to include, at a minimum, backup electrical power.

Note: Most LPOE have backup generators that should be able to provide backup power to the command center. Any backup power requirements will be discussed with GSA.

10 Human Systems Integration (HSI) Requirements

HSI requirements include human factors requirements, such as human-machine interface or ergonomic requirements.

10.1 LEP Unit HSI Requirements

- 10.1.1 The LEP System shall provide audio and visual cues in and around the scan area when the LEP Units are in scanning (X-ray transmit) mode.
- 10.1.2 The LEP System shall provide audio and visual cues to the operator when the LEP Units are in scanning (X-ray transmit) mode.

10.2 Work Center HSI Requirements

- 10.2.1 The work center shall turn-off X-ray upon operator command.

11 Environmental, Safety, and Occupational Health (ESOH) Requirements

ESOH requirements describe what is necessary to comply with existing ESOH policies or regulations (e.g., National Environmental Policy Act [NEPA], Occupational Safety and Health Act [OSHA], Clean Air Act [CAA]) and minimize environmental, safety, and occupational health risks.

11.1 LEP System ESOH Requirements

- 11.1.1 The LEP System shall maintain fully operability in the environmental extremes seen on the northern and southern border without degradation in its intended outdoor environment.
- 11.1.2 The LEP System shall include safety interlocks such as vehicle proximity/presence detection for all systems for the protection of pedestrian traffic and untrained personnel.
- 11.1.3 The LEP System shall operate under environmental conditions along the northern and southern borders as specified in the table below LEP System Environmental Conditions.

Table 8 LEP System Operational Environment

#	Category	Requirement
1.	Relative Humidity	Up to 98% RH

#	Category	Requirement
2.	Moisture and Dust Protection	Comply with IP code 54 (see IEC 60529), the ingress of dust and splashing water.
3.	Temperature Range	Extreme temperature ranges from -40°C to +60 °C (-40°F to 140 °F).
4.	Wind	• Surface average wind speeds up to 60 mph with gusts up to 75 mph. • Survive wind speeds of up to 125 mph.
5.	Precipitation	Up to 2 inches/hr
6.	Electromagnetic Interference	N42.41-2007 requirements in Sections 6.3-6.8 and 6.10 (for outdoor use)
7.	Salt Fog	5% by weight. Reference MIL-STD-810, Method 509.4

11.1.4 The LEP System shall include individual Emergency Stop Switches for all systems in the control center at the LEP operator position and on the LEP systems themselves, outside of the Radiation Control Area (RCA) to manually interrupt the scanning process by disabling the transmission of X-rays in an emergency or when there is an unauthorized entry inside the system's RCA in accordance with ANSI N43.3-2008, paragraph 7.05.2, and ANSI N43.14-2011.

11.1.5 The LEP System shall include devices (bollards, guard rails, etc.) to prevent/protect the system and associated subsystem/components from being hit by vehicles traveling through/near the system.

11.1.6 The LEP System shall have guard rails that ensure all vehicles with an outer track width (i.e., the dimension from the outer tire tread on one tire to the outer tire tread on the opposite tire on the same end of the vehicle) equal to or less than the width of the scanning area are able to pass through the system with clearance of 6" (i.e. if the scanning area is 8' 6" wide, then guard rails are at least 9' wide).

11.1.7 The system shall illuminate the vehicle within the tunnel and operating area.

11.2 LEP Unit X-Ray System ESOH Requirements

- 11.2.1 The LEP Unit shall not rely on radioactive isotopes.
- 11.2.2 The LEP Unit shall perform X-ray transmission and / or backscatter scan of the vehicle at low energy / flux allowing a safe exposure rate.
- 11.2.3 The radiation dose to vehicle occupants in adjacent lanes shall not exceed 50 μ R in any one hour.
- 11.2.4 The LEP Unit shall have a controlled RCA which limits the potential exposure to personnel outside of this area to < 50 μ R in any one hour.
- 11.2.5 Engineering controls such as barricades or fencing shall be installed to maintain the controlled RCA
- 11.2.6 The LEP System shall stop scanning, if the subject vehicle speed is less than 2 mph and annotate the data package.
- 11.2.7 The LEP System, plus UVBS if equipped, shall limit radiological dose equivalent to each vehicle occupant being scanned to no more than 25 μ R/scan (T) .

Note: Requirement must be met at >2 mph and <10 mph scan speed and analysis includes adjacent lane vehicles and impact of traffic queue delays on LEP radiation exposure to drivers or passengers, per ANSI 43.17 2009.

- 11.2.8 The LEP System, plus UVBS if equipped, should limit radiological dose equivalent to each vehicle occupant being scanned to no more than 10 μ R /scan (O).

Note: Requirement must be met at >2 mph and <10 mph scan speed and analysis includes adjacent lane vehicles and impact of traffic queue delays on LEP radiation exposure to drivers or passengers, per ANSI 43.17 2009.

- 11.2.9 Approval from the CBP Radiation Safety Officer (RSO) shall be obtained for installations that do not have X-ray opt-out lane(s).

11.3 Traffic Control System (TCS) ESOH Requirements

- 11.3.1 A hard barrier capable of withstanding a low speed ($\leq$ 15 mph) strike shall be placed immediately prior to the entrance of the LEP as a safeguard.

11.4 CCTV and Camera Systems ESOH Requirements

- 11.4.1 The CCTV System shall provide operators with full situational awareness for the LEP System and adjacent operations.

12 Interoperability Requirements

Interoperability requirements describe the parameters that must be a part of, or met by, the system to interoperate with other systems or components (e.g., ability to exchange information and to use the information exchanged).

12.1 LEP System Interoperability Requirements

- 12.1.1 The LEP System shall be able to operate independently of all other lanes.

12.2 LEP Unit X-Ray System Interoperability Requirements

- 12.2.1 The System shall not interfere with RPM gamma-ray detection operations.
- 12.2.2 The System shall employ a pulsed photon generator or other similar technology (linac, betatron or similar) to facilitate RPM blanking.
- 12.2.3 The System shall produce a compatible blanking signal, or other similar technology, with adjustable timing which can be transmitted to any RPMs in proximity (see 8.2.1).
- 12.2.4 The blanking signal, or other similar technology, shall remain active regardless of LEP operational status.
- 12.2.5 The system shall be capable of meeting the requirements in this standard while operating to integrate with existing blanking infrastructure.
- 12.2.6 If multiple NII systems are deployed in close proximity, the photon pulse frequency, or similar appropriate technology, from each system shall be synchronized and produce a single blanking signal to interface with RPMs in proximity.
- 12.2.7 The System shall not interfere with RPM neutron detection operations.
- 12.2.8 Any neutron blanking signal should be delivered independently of the gamma blanking signal to minimize the neutron blanking duration (O).
- 12.2.9 Systems shall be designed and shielded to reduce the production of neutrons to the level of background at any RPMs in proximity (T).
- 12.2.10 Systems should be designed and shielded to reduce the production of neutrons to the level of background at 100 feet in all directions from the system center (O).

- 12.2.11 Systems shall operate without interference from collocated NII systems of the same model with greater than 3 ft (T) separation of outer boundaries
- 12.2.12 Systems should operate without interference from collocated NII systems of the same model with greater than 1 ft (O) separation of outer boundaries.

12.3 Under-Vehicle Back-Scatter System (UVBS) Objective - Interoperability Requirements

- 12.3.1 If included, the UVBS shall be fully integrated with and use the LEP Unit LPR system.

12.4 Vehicle Detection Interoperability Requirements

- 12.4.1 Each LEP Unit 'lane' vehicle presence detection shall be integrated with the traffic control system.

12.5 CCTV and Camera Systems Interoperability Requirements

- 12.5.1 The Fixed CCTV system shall be independent of the PTZ cameras at the entrance and exit of the LEP Units

12.6 LEP Work Center Interoperability Requirements

- 12.6.1 Each LEP Work Center shall allow any analyst to adjudicate any X-ray image from any of the connected LEP Units.

13 Security & Privacy Requirements

Security and privacy requirements describe the need to protect data and information (e.g., screening images) from intrusion, corruption, repudiation, theft, unauthorized access and control of access to the data. This includes controlling who may view and alter application data. Specify any requirements regarding security or privacy issues surrounding use of the product or protection of the data used or created by the product.

13.1 LEP System Security & Privacy Requirements

- 13.1.1 Audit Record Retention – Audit records shall be maintained on the system for a minimum of 30 days.
- 13.1.2 Audit Record Retention - Audit records shall be preserved for a period of 7 years as part of managing records for each system to allow audit information to be placed online for analysis with reasonable ease.
- 13.1.3 Audit Reduction / Report Generation – The system shall provide an audit reduction and report generation capability that supports on-demand audit review, analysis, and reporting requirements

and after-the-fact investigations of security incidents without altering the original content or time ordering of audit records

Note: DHS 4300A policy requires audit record generation capability for relevant or auditable events identified by DHS. National Institute of Standards and Technology (NIST) defines a relevant event as an occurrence (e.g., an auditable event or flag) considered to have potential security implications to the system or its environment that may require action.

13.2 Data Storage (DS) Security & Privacy Requirements

13.2.1 The LEP System DS shall store data in encrypted form.

13.3 LEP Work Center Security & Privacy Requirements

13.3.1 The LEP workstations shall be compliant with the latest CBP Technical Reference Model (TRM) specification.

13.4 Access Control Security & Privacy Requirements

13.4.1 Unique User Accounts – The system shall enable all users to have unique account identifiers, including separate identifiers for operator accounts and administrator accounts.

13.4.2 Automatic Account Lockout – The system shall lock a user account for twenty (20) minutes after three consecutive failed logon attempts

13.4.3 Automatic Session Termination – The system shall lock the user login session after twenty (20) minutes of inactivity.

13.4.4 Automatic Session Termination – The system shall require the user to authenticate with the user account password to regain system access after the system lock.

13.4.5 Automatic Session Termination – After 60 minutes of inactivity, the system shall log the user account out of the system.

13.4.6 Warning Banner – The DHS Chief Information Security Officer (CISO) stipulates that a warning banner statement be displayed on all DHS systems during user account logon. The system shall prompt the user to accept the current language before they can authenticate with the user id and user password to access the system. The current language required follows: 1. You are accessing a U.S. Government information system, which includes (1) this computer, (2) this computer network, (3) all computers connected to this network and (4) all devices and storage media attached to this network or to a computer on this network. This information system is provided for U.S. Government-authorized use only. 2. Unauthorized or improper use or access of this

system may result in disciplinary action, as well as civil and criminal penalties. 3. By using this information system, you understand and consent to the following: a. You have no reasonable expectation of privacy when you use this information system; this includes any communications or data transiting, stored on, originated from or directed to this information system. At any time, and for any lawful Government purpose, the Government may monitor, intercept, search and seize any communication or data transiting, stored on, originated from or directed to or from this information system. b. The Government may disclose or use any communications or data transiting, stored on, originated from or directed to or from this information system for any lawful Government purpose. c. You are NOT authorized to process classified information on this information system.

13.5 Auditing Security & Privacy Requirements

- 13.5.1 The System shall retain audit records sufficient in detail to facilitate the reconstruction of events if compromise or malfunction occurs.
- 13.5.2 Audit records shall contain at least the following information: 1. Identity of each user and device accessing or attempting to access the system. 2. Time and date of the access and the logoff. 3. Activities that might modify, bypass, or negate information security safeguards. 4. Security – relevant actions associated with processing. 5. All activities performed using an administrator account/identity.
- 13.5.3 Audit Record Access – Audit records and audit logs shall be protected from unauthorized access, modification, or destruction.

13.6 Computer Security & Privacy Requirements

- 13.6.1 System security shall follow DHS/CBP Enterprise and Security Architecture requirements for a system categorized as Moderate/Low under FIPS-199 and collecting privacy sensitive information.
- 13.6.2 The System shall be compliant with DHS Sensitive Systems Policy Directive 4300A, Version 13.2.
- 13.6.3 The System shall be compliant with CBP Information Systems Security Policies and Procedures, HB 1400-05D, for access control (latest version or superseding document).
- 13.6.4 The System work station shall disable wireless communications, upon administrator command

- 13.6.5 The System shall comply with the DHS hardening guides for operating systems
- 13.6.6 The System shall comply with the DHS configuration guides for applications.
- 13.6.7 The System DS shall encrypt in accordance with FIPS 197 Advanced Encryption Standard (AES), Chapter 5, algorithms.
- 13.6.8 The System shall encrypt with at least 256 bit in accordance with FIPS 140-2 Standards for Security Requirements for Cryptographic Modules.
- 13.6.9 Desktop computers shall be equipped with operating systems of Windows 11 or greater that are on the CBP TRM.
- 13.6.10 Basic Requirements - Any legacy air gapped non-integrated systems shall conform with the following requirements:
 - 13.6.10.1 Use upgradable operating system software and approved applications
 - 13.6.10.2 Not use or disable WiFi
 - 13.6.10.3 Use operating systems that meet the requisite OHS STIGs for hardening configurations
 - 13.6.10.4 Use required encryption for data at rest
 - 13.6.10.5 Be cyber resilient to data manipulation, unauthorized download and non-malicious incidents
- 13.6.11 Integrated networked systems shall follow the requirements of non-integrated systems and additionally:

- 13.6.11.1 Only use WIFI designs which are explicitly reviewed and approved by the CBP EIOD organization
- 13.6.11.2 Only use other wireless connectivity as approved by Enterprise Infrastructure and Operations Directorate (EIOD)
- 13.6.11.3 Be patched and maintained according to OHS patching standards (within 30 days usually)
- 13.6.11.4 Not allow Username/Password authentication
- 13.6.11.5 Require Personal Identity Verification (PIV) authentication
- 13.6.11.6 Integrate with CBP identify management lifecycle systems
- 13.6.11.7 Integrate with enterprise audit log repository (SPLUNK)
- 13.6.11.8 Be capable of receiving operating system security patches and updates and leverage CBP enterprise security notification services for network security alert monitoring

13.7 Identification and Authentication Security & Privacy Requirements

- 13.7.1 Levels of Access – The System shall provide three levels of CBP User access (Operator Level 1, Operator Level 2, and Administrator).
- 13.7.2 LEPs not integrated / connected with the CBP network shall comply with the following. Passwords – All passwords used to access the NII system for operation shall be required to be Strong Passwords. Strong passwords have the following requirements:
 - 1) Are at least 12 characters in length.
 - 2) Comply with the DHS hardening guides for operating systems and the configuration guides for applications. In the absence of guidance, the ISSO will determine the appropriate password complexity based on the level of risk.
 - 3) The system shall force all passwords to contain three of the four following types of characters.
 - i. Uppercase Alphabetic characters
 - ii. Lowercase Alphabetic characters
 - iii. Special characters
 - iv. Numbers
 - 4) The system shall check that the passwords are not the same as any of the user's previous 8 passwords when creating passwords. If they are the same as any of the user's previous 8 passwords the

system shall force the user to change to a password that is not the same as any of the previous eight passwords.

- i. Strong passwords are not the same as the User identification (ID).
 - ii. Strong passwords shall contain special characters.
- 13.7.3 Strong pass phrases, if used in addition to or instead of passwords, are forced to follow the same guidelines as above.
- 13.7.4 The system shall deactivate unused accounts automatically if they have not been used in 45 days.
- 13.7.5 Passwords shall not be set to the null string.
- 13.7.6 Automatic passwords shall change frequency. The system shall require the user to change their password every 90 days.
- 13.7.7 Password Storage – The system shall store all user account passwords in encrypted form in accordance with DHS/CBP encryption requirements.

13.8 Printing Security & Privacy Requirements

- 13.8.1 The System documents shall be printed with "FOR OFFICIAL USE ONLY - LAW ENFORCEMENT SENSITIVE" in the header or footer.
- 13.8.2 The footer of printed documents shall indicate system type and location.

14 Reliability, Availability, and Maintainability (RAM)

The contractor shall design, engineer, manufacture, and deploy the system and sub-systems to achieve established outcomes and ensure the availability of the capability throughout its intended service life expectancy by applying the tenets of CBP's Cost Wise Readiness (CWR) metrics. Accordingly, the Contractor shall sustain the system and/or key subsystems during the warranty period in a manner that meets the Operational Availability (Ao) service requirements. The Contractor is encouraged to conduct an analytical assessment that offers an alternative balanced solution for optimizing reliability and downtime to achieve readiness.

The Contractor will address the Reliability, Availability, and Maintainability (RAM) in the Lifecycle Sustainment and Support Requirements attachment within this Statement of Work.

Remainder of page left intentionally blank